Finite spectral evaluation and uniform filling bounds

Abstract

We study a finite functional on polynomials of degree less than N, whose spectral nodes are the roots of $2T_N(z/2)+2$ and whose multiplier is an integer Gaussian chirp. Double roots retain first derivatives; the odd simple root has its own value term. We prove that the functional has supremum-norm bound $1+|n|/2$, independently of the level N. If the polynomial vanishes at every node, the bound improves to $(N-1)/N$, independently also of the signed integer n. Tensor products admit a further exact subtraction when only joint grid vanishing is known. The proof uses canonical polynomial remainders, a finite midpoint identity, and an explicit derivative interpolation formula. We also construct a Fourier similarity between a concrete pair Casimir and its polynomial face operator, so the resolvent reproduction identity holds on the full matrix, including nilpotent parts. Regular confluent specialization identifies the Gaussian multiplier with powers of an explicit finite crossing matrix. For a planar diagram obtained from a matched triangulated sphere, we prove that its reduced tensor evaluation is one coefficient of the product of its local face polynomials. The proof retains the actual cups, caps and marked cut, including the shifted face lattice at even levels. An explicit matrix calculation identifies the resulting knot evaluation with the reduced Kashaev invariant, up to its stated unit framing factor. No volume asymptotic is assumed or proved.

1 Introduction and the main estimate

At a repeated spectral root, a polynomial is not determined by its values. Its derivative can act on a nonzero nilpotent part of an operator. This elementary fact matters when a crossing power is inserted into a finite spectral source: a value-only sum would omit some of the original matrix entries. The purpose of this paper is to retain those entries while bounding the insertion by the supremum of the original polynomial on a real interval. We then prove the exact diagram identity needed to apply that bound: the face variables shared by different crossings are identified before taking the prescribed coefficient.

Fix $N \geq 2$, and write

$$d = N-1, \qquad q = e^{\pi i/N}, \qquad Q = q^2, \qquad \chi_N(z) = 2T_N(z/2)+2.$$

The Chebyshev polynomial is defined by $T_N(\cos\phi) = \cos(N\phi)$; χ_N is monic of degree N. Its distinct double roots are

$$c_a = 2\cos\frac{(2a+1)\pi}{N}, \qquad 0 \leq a < \lfloor N/2 \rfloor.$$

For odd N there is one further root, -2, of multiplicity one. All statements below apply to both parities and to composite levels.

Let $\mathcal{R}_{\chi_N} F$ be the canonical polynomial remainder modulo χ_N, and put

$$\ell_N(F) = [z^{N-1}]\mathcal{R}_{\chi_N} F.$$

For $n \in \mathbb{Z}$, the multiplier f_n is specified at the spectral roots by the local germs

$$f_n(2\cos\phi) = q^{nd^2/4} \exp\left(\frac{inN(\phi-\pi)^2}{4\pi}\right), \qquad 0 < \phi < 2\pi.$$

The formula is symmetric about π and defines a regular germ at $z = -2$. Its Hermite polynomial $\widetilde{f}_n$ is the unique polynomial of degree less than N with those values and first derivatives at the double roots, and the specified value at the odd simple root. Define

$$\mathcal{L}_N^n H = \ell_N(\widetilde{f}_n H).$$

The product is taken before reduction. In particular ℓ_N is not being treated as a multiplicative functional.

Theorem 1.1 (Uniform finite filling bound)**.** *For every complex polynomial H of degree at most d and every signed integer n,*

$$|\mathcal{L}_N^n H| \le (1 + |n|/2)\|H\|_{[-2,2]}. \tag{1.1}$$

If H vanishes at all distinct roots of χ_N, then

$$|\mathcal{L}_N^n H| \le (d/N)\|H\|_{[-2,2]}. \tag{1.2}$$

For a polynomial S in r spectral variables, of degree at most d in each variable, suppose that it vanishes whenever σ_e is a spectral node, with the other variables arbitrary, for each e in a set J. Then

$$\left|\left(\bigotimes_{e=1}^{r} \mathcal{L}_N^{n_e}\right) S\right| \le (d/N)^{|J|} \prod_{e \notin J} (1 + |n_e|/2)\, \|S\|_{[-2,2]^r}. \tag{1.3}$$

Finally, if $\mathcal{D}$ has the same degree bounds and vanishes only on the Cartesian product of all node sets, then

$$\left|\left(\bigotimes_{e=1}^{r} \mathcal{L}_N^{n_e}\right) \mathcal{D}\right| \le \left\{\prod_e (1 + |n_e|/2) - \prod_e (1/N + |n_e|/2)\right\} \|\mathcal{D}\|_{[-2,2]^r}. \tag{1.4}$$

Here $\|H\|_{[-2,2]} = \max_{-2\le x\le 2} |H(x)|$, and the cube norm has the analogous meaning. The tensor product denotes successive application in the separate spectral variables. These operations commute because they act in different scalar variables; this says nothing about commutation of matrices at different crossings.

The derivative inequality used below is the classical Bernstein inequality, obtained here from Riesz's finite interpolation formula [2]. The interpolation and polynomial-division tools are standard. The calculations in this paper specify their application to the stated Gaussian Hermite functional, with its normalization and tensor-grid subtraction.

The proof of the theorem occupies Sections 3–8. Its essential estimate is a finite derivative formula with total coefficient mass d. Thus the constant is independent of the number of

interpolation nodes, even though the functional retains derivatives at nearly every distinct root. The degree restriction is essential to this argument and will be retained in every application.

The knot-theoretic motivation is the relation between root-of-unity quantum invariants and hyperbolic volume initiated by Kashaev [3] and Murakami–Murakami [4]. The present theorem is a finite estimate. It separates the cost of inserting fixed crossing powers from the harder problem of estimating a shared source polynomial. Section 9 proves the global coefficient identity, including the turns, the reduced mark and the face lattice. Section 10 compares its literal crossing matrices and normalization with the reduced Kashaev invariant. The last section states the remaining asymptotic implication: a nonzero leading contribution and a strict bound for the entire remainder give the rate of the full invariant. Those analytic estimates and the identification of their rate with hyperbolic volume are not proved here.

2 Polynomial reduction and resolvent reproduction

The value-and-derivative interpolation is the classical construction of Hermite [1]. We record its elementary polynomial form, including repeated roots, because it fixes the coefficient functional used here. Before using the particular spectral roots, we record the algebraic fact behind reduction at repeated roots. It explains which parts of the calculation depend only on the polynomial and which still require the original crossing operator.

Let R be a commutative ring with identity, and let $r_1, \dots, r_m \in R$, with $m \geq 1$. Repetitions are allowed. Put $p(X) = \prod_{j=1}^{m}(X - r_j)$. Given $f \in R[X]$, define $f_0 = f$ and successively define f_j by

$$f_{j-1}(X) = f_{j-1}(r_j) + (X - r_j)f_j(X), \qquad 1 \leq j \leq m.$$

Each f_j is obtained by synthetic division. This is a polynomial operation: it uses no inverse in R, even if two roots coincide. Define

$$\mathcal{R}_p f(X) = \sum_{j=1}^{m} f_{j-1}(r_j) \prod_{i<j}(X - r_i), \qquad \ell_p(f) = [X^{m-1}]\mathcal{R}_p f.$$

Here $[X^k]$ means the coefficient of X^k, and an empty product is one.

Proposition 2.1 (Canonical polynomial remainder)**.** *The polynomial $\mathcal{R}_p f$ is the unique polynomial of degree less than m for which $f - \mathcal{R}_p f$ is a multiple of p. Consequently it is independent of the ordering of the root list, including repeated entries. Both $\mathcal{R}_p$ and ℓ_p are R-linear, and adding a multiple of p to their input leaves their values unchanged. Moreover,*

$$\ell_p(f) = f_{m-1}(r_m).$$

Among additive functionals on $R[X]$, ℓ_p is uniquely specified by the two requirements

$$F(pq) = 0 \quad \textit{for every } q \in R[X], \qquad F(r) = [X^{m-1}]r \quad \textit{whenever } \deg r < m.$$

Proof. Substituting each synthetic-division identity into the preceding one gives $f = pf_m + \mathcal{R}_p f$. Every summand in the displayed remainder has degree at most $m - 1$. If two such remainders existed, their difference would have degree less than m and would be a multiple of the monic polynomial p. A nonzero multiple pq has degree $m + \deg q$: its leading coefficient is the leading

coefficient of q, because that of p is one. This argument works even when R has zero divisors. The difference must therefore vanish.

Uniqueness proves independence of root order, linearity and invariance under adding a multiple of p. In the Newton expression for the remainder, only the last summand can contribute to X^{m-1}; its product of linear factors is monic. This proves the formula for ℓ_p. Finally, any additive F with the two stated properties satisfies

$$F(f) = F(pf_m) + F(\mathcal{R}_p f) = [X^{m-1}]\mathcal{R}_p f = \ell_p(f).$$

□

For two equal roots this construction gives

$$\mathcal{R}_{(X-r)^2} f(X) = f(r) + f'(r)(X-r), \qquad \ell_{(X-r)^2}(f) = f'(r).$$

The derivative is the formal polynomial derivative. Thus repeated roots retain derivative information; replacing them by a single evaluation changes the operation.

Proposition 2.2 (Products and repeated-root derivatives)**.** *For any $f, g \in R[X]$,*

$$\mathcal{R}_p(fg) = \mathcal{R}_p\big((\mathcal{R}_p f)(\mathcal{R}_p g)\big), \qquad \ell_p(fg) = \ell_p\big((\mathcal{R}_p f)(\mathcal{R}_p g)\big).$$

Consequently a finite product may be reduced after each multiplication, with the same final remainder and coefficient functional. This reduced multiplication is associative. If r occurs at least twice among the roots of p, the final remainder retains both $(fg)(r)$ and

$$(fg)'(r) = f'(r)g(r) + f(r)g'(r).$$

For the modulus $(X-r)^2$ its full expression is

$$\mathcal{R}_{(X-r)^2}(fg) = f(r)g(r) + \big(f'(r)g(r) + f(r)g'(r)\big)(X-r).$$

No degree bound on the unreduced factors or their product is required.

Proof. Write $f = \mathcal{R}_p f + pa$ and $g = \mathcal{R}_p g + pb$ using synthetic division. Their product differs from $(\mathcal{R}_p f)(\mathcal{R}_p g)$ by $p\{a\mathcal{R}_p g + b\mathcal{R}_p f + pab\}$. Uniqueness of the remainder proves the first equality; taking its top coefficient proves the second. Repeating the equality proves the finite-product statement. Either parenthesization of a product of three factors reduces the same polynomial fgh, which proves associativity.

When $(X-r)^2$ divides p, the difference between any polynomial and its remainder has both value and formal derivative zero at r. The formal product rule gives the displayed derivative. The explicit degree-one remainder then follows from the double-root formula above. □

The coefficient functional is linear, but it is generally not multiplicative: $\ell_p(fg)$ need not equal $\ell_p(f)\ell_p(g)$. The product must be formed and reduced before its coefficient is taken. At a repeated spectral root the product rule also retains the first derivative of either factor; a list of values alone cannot replace it. This permits a product of arbitrary degree to be formed first and reduced only when the coefficient functional is applied.

There is also a direct algebraic relation between this coefficient functional and the resolvent numerator. It does not require a diagonalization of the operator.

Proposition 2.3 (Reproduction by the polynomial resolvent)**.** *Let $p(X)$ be a split monic polynomial of degree $m \geq 1$ over a commutative algebra A, and let $w \in A$ satisfy $p(w) = 0$. Define V_w by polynomial synthetic division,*

$$V_w(X) = \frac{p(X) - p(w)}{X - w}.$$

The fraction denotes a polynomial, with no inverse assumed in A. Let ℓ_p be the coefficient of X^{m-1} in the canonical remainder modulo p. For every $f \in A[X]$,

$$\ell_p(fV_w) = f(w).$$

Repeated roots of p are allowed.

Proof. Synthetic division gives $(X - w)V_w = p$. The polynomial V_w is monic of degree $m - 1$, so $\ell_p(V_w) = 1$. For every polynomial g, the identity $(X - w)gV_w = pg$ and the vanishing of ℓ_p on multiples of p imply

$$\ell_p(XgV_w) = w\ell_p(gV_w).$$

Induction gives $\ell_p(X^jV_w) = w^j$. Linearity proves the claim for arbitrary f, with no restriction on its degree. □

For a matrix application, use the commutative algebra generated by one matrix Ω and scalar matrices. If $p(\Omega) = 0$, then the proposition applies with $w = \Omega$ and gives the full matrix $f(\Omega)$, including its nilpotent part. No diagonalization is required. A concrete Casimir matrix satisfying the required relation is treated in Section 4.

3 Repeated roots, the odd simple root, and the midpoint identity

For completeness, the recurrence $2T_{j+1}(z/2) = z\,2T_j(z/2) - 2T_{j-1}(z/2)$ shows that χ_N is monic of degree N. In the variable ϕ, $\chi_N(2\cos\phi) = 2\cos(N\phi) + 2$ has a double zero whenever $N\phi$ is an odd multiple of π. Away from $\phi = \pi$ the change of variable has nonzero derivative, so these give double roots in z. For odd N, near $\phi = \pi + t$ one has $\chi_N(2\cos\phi) = N^2t^2 + O(t^4)$ and $2\cos\phi + 2 = t^2 + O(t^4)$, giving the simple root -2. The resulting multiplicities add to N, so there are no others.

Write

$$c_a = 2\cos\frac{(2a+1)\pi}{N}, \qquad 0 \leq a < \lfloor N/2 \rfloor. \tag{3.1}$$

These are the distinct double roots. The only remaining root is -2, simple, when N is odd. At a double root c, put $h(z) = \chi_N(z)/(z - c)^2$. The Chebyshev differential equation gives

$$h(c) = \frac{N^2}{4 - c^2}, \qquad \frac{h'(c)}{h(c)} = \frac{c}{4 - c^2}; \qquad \chi_N'(-2) = N^2 \quad (N \text{ odd}). \tag{3.2}$$

Indeed $(4 - z^2)\chi_N'' - z\chi_N' + N^2(\chi_N - 2) = 0$. Evaluate it and its derivative at c, using $\chi_N(c) = \chi_N'(c) = 0$, to obtain (3.2).

The formula for the coefficient functional follows directly from partial fractions. For a polynomial F, replace it by its remainder $r = \mathcal{R}_{\chi_N} F$, of degree less than N. The coefficient of z^{-1} in $r(z)/\chi_N(z)$ at infinity is $\ell_{\chi_N}(F)$. At a double root c, the coefficient of $(z-c)^{-1}$ is

$$\left(\frac{r}{h}\right)'(c) = \frac{r'(c)}{h(c)} - \frac{h'(c)r(c)}{h(c)^2}.$$

At the odd simple root it is $r(-2)/\chi_N'(-2)$. Summing these coefficients and using (3.2) gives

$$\mathcal{L}_N^f H = \sum_a \frac{(4-c_a^2)(fH)'(c_a) - c_a f(c_a)H(c_a)}{N^2} + \mathbf{1}_{N \text{ odd}} \frac{f(-2)H(-2)}{N^2}. \tag{3.3}$$

Here f may be a specified holomorphic germ at each root. Its Hermite polynomial is the unique polynomial of degree less than N with those values and first derivatives at the double roots, and its value at the odd simple root. Existence follows because the corresponding linear map between two N-dimensional spaces has zero kernel: a polynomial with all these jets zero is divisible by the monic degree-N polynomial χ_N. We define $\mathcal{L}_N^f H = \ell_{\chi_N}(\widetilde{f}H)$, where $\widetilde{f}$ is that Hermite polynomial. Formula (3.3) therefore applies to every polynomial H, without choosing a global branch for f. Its prime denotes differentiation in z.

Let

$$\phi_k = \frac{(2k+1)\pi}{N}, \quad 0 \le k < N, \qquad P(\phi) = H(2\cos\phi). \tag{3.4}$$

Each double root appears twice in this grid. At the odd simple root, $\phi = \pi$ occurs once, and differentiating the sine supplies exactly its weight in (3.3). Thus (3.3) is equivalently

$$\mathcal{L}_N^f H = -\frac{1}{2}\sum_{k=0}^{N-1} \frac{d}{d\phi} \frac{2\sin\phi\, f(2\cos\phi)H(2\cos\phi)}{N^2}\bigg|_{\phi=\phi_k}. \tag{3.5}$$

There is no omission or doubling of the odd simple node.

For the integer n in Theorem 1.1, the centered Gaussian is

$$f_n(2\cos\phi) = q^{nd^2/4} \exp\left(\frac{inN(\phi-\pi)^2}{4\pi}\right), \qquad 0 \le \phi \le 2\pi. \tag{3.6}$$

Here q^a means $e^{\pi i a/N}$, fixing the original square-root lift even for nonintegral a. In particular

$$|f_n| = 1, \qquad |\partial_\phi f_n| \le \frac{|n|N}{2}, \qquad f_n(c_a) = q^{nd^2/2}(-1)^{an} q^{na(a+1)}. \tag{3.7}$$

The phase in this formula is retained in the functional; it is not replaced by its absolute value. Formula (3.6) is symmetric about π; it has the required holomorphic germ at the odd simple node. The possible endpoint cut at $z = 2$ does not occur at a spectral node and does not enter the finite identity (3.5).

The germ at $z = -2$ is regular. Indeed, with $t = \phi - \pi$ one has $z = -2\cos t = -2 + t^2 + O(t^4)$, so t^2 is a holomorphic function of $z+2$. At a double root put $h = 2a+1 < N$. Differentiating the Gaussian and using $dz/d\phi = -2\sin\phi$ gives the exact normal ratio

$$\frac{f_n'(c_a)}{f_n(c_a)} = -\frac{n(N-h)}{2(q^h - q^{-h})}. \tag{3.8}$$

Expanding the square at $\phi = h\pi/N$ gives the value in (3.7); the discarded difference in its exponent is an integer multiple of $2\pi i n$. This calculation is valid for every integer n, positive, negative, or zero, and every $N \geq 2$.

For comparison, let r_N be the degree-less-than-N polynomial with

$$r_N(c_a) = (-1)^a q^{a(a+1)}, \qquad \frac{r_N'(c_a)}{r_N(c_a)} = -\frac{N-1-2a}{2(q^{2a+1}-q^{-2a-1})},$$

and, when $N = 2m+1$, the value $r_N(-2) = (-1)^m q^{m(m+1)}$ at the simple root. All its nodal values are nonzero, so its class is a unit in $\mathbb{C}[z]/(\chi_N)$. The product rule, including negative powers of a unit, and (3.8) prove

$$\widetilde{f}_n = q^{nd^2/2} r_N^n \pmod{\chi_N}. \tag{3.9}$$

This is an identity of values and normals, not merely an equality of functions on the set of distinct roots. Section 5 identifies $q^{d^2/2} r_N(\Omega)$ with an explicitly defined finite crossing. The norm proof itself requires only the displayed jets.

For the next formula, write $f_k = f_n(2\cos\phi_k)$ and $P_k = P(\phi_k)$. Here f_k' is the derivative of $\phi \mapsto f_n(2\cos\phi)$ at ϕ_k, and $P_k' = P'(\phi_k)$. Expanding (3.5) gives

$$\mathcal{L}_N^n H = -\frac{1}{N^2} \sum_{k=0}^{N-1} (\cos\phi_k f_k P_k + \sin\phi_k f_k' P_k + \sin\phi_k f_k P_k'). \tag{3.10}$$

This finite formula is sufficient for every norm below; no interchange of an infinite Fourier series or a contour limit is needed.

4 The local source identity from a finite Fourier transform

We give the local matrix calculation, including its normalization. It supplies an explicit matrix application of the polynomial functional and a homogeneous four-variable representation of its entries. No decomposition into simple modules is needed.

Put $Q = q^2$. In the balanced basis $e_0, \ldots, e_d$ of the module of highest weight d and dimension N, the matrices are

$$Ke_i = q^{d-2i} e_i, \qquad Fe_i = (1-Q^{i+1}) e_{i+1}, \qquad \mathcal{E} e_i = (Q^{-i}-1) e_{i-1}.$$

Here $\mathcal{E} = (q-q^{-1})^2 E$, and a term outside $0, \ldots, d$ is zero. One can obtain these matrices from the basis v_i with $Ev_i = [i]_q^2 v_{i-1}$ and $Fv_i = v_{i+1}$ by setting $e_i = v_i/(Q;Q)_i$. Here $[i]_q = (q^i - q^{-i})/(q-q^{-1})$ and $(Q;Q)_i = \prod_{j=1}^{i}(1-Q^j)$, with the empty product equal to one. All denominators are nonzero because $i < N$. Alternatively the displayed matrices can be taken as the definition of the model. The pair matrices are

$$\begin{aligned} K_{12} &= K \otimes K, \\ F_{12} &= F \otimes I + K^{-1} \otimes F, \\ \mathcal{E}_{12} &= \mathcal{E} \otimes K + I \otimes \mathcal{E}, \\ \Omega &= F_{12}\mathcal{E}_{12} + qK_{12} + q^{-1}K_{12}^{-1}. \end{aligned}$$

On an input $e_u \otimes e_v$, set $U = Q^u, V = Q^v$. Multiplying these displayed matrices gives the diagonal coefficient D, and the two coefficients A_+, A_- that change the first index by $+1, -1$:

$$\begin{aligned} D(U,V) &= 2qU + 2q^{-1}V^{-1} - (q+q^{-1})U/V, \\ A_+(U,V) &= (V^{-1}-1)(1-QU), \\ A_-(U,V) &= (Q^{-1}V^{-1}-1)(1-U). \end{aligned} \tag{4.1}$$

The second index changes by the opposite amount. The positive shift vanishes when $u = d$ or $v = 0$, and the negative shift vanishes when $u = 0$ or $v = d$. Thus these formulas retain the true integer total and all physical endpoints.

For independent outer face variables x, y, define an operator on polynomials in z by

$$\begin{aligned} (Tg)(z) = &-q^{-1}\left(1 + \frac{q(x+y)}{z} + \frac{q^2xy}{z^2}\right) g(Q^{-1}z) \\ &- q\left(1 + \frac{q^{-1}(x+y)}{z} + \frac{q^{-2}xy}{z^2}\right) g(Qz) \\ &+ \left(\frac{2(x+y)}{z} + \frac{(q+q^{-1})xy}{z^2}\right) g(z). \end{aligned} \tag{4.2}$$

Its denominators cancel on polynomials. Indeed,

$$Tz^j = A_j z^j + B_j(x+y)z^{j-1} + C_j xy z^{j-2},$$

where

$$\begin{aligned} A_j &= -(qQ^j + q^{-1}Q^{-j}), \\ B_j &= 2 - Q^j - Q^{-j}, \\ C_j &= q + q^{-1} - qQ^{-j} - q^{-1}Q^j. \end{aligned}$$

The only potentially negative powers have coefficients $B_0 = C_0 = C_1 = 0$. In particular T preserves the N-dimensional space of polynomials of degree at most d, and it preserves total degree when x, y, z all have degree one.

Proposition 4.1 (The actual Casimir and the face operator)**.** *For $s = 0, \dots, d$, let Ω_s be the restriction of Ω to the ordered basis*

$$e_u \otimes e_{[s-u]_N}, \qquad 0 \le u < N,$$

where $[a]_N$ is the representative of a in $\{0, \dots, N-1\}$. Choose $y \neq 0$, set $x = Q^{s+1}y$, and put

$$b_u = -q^{-1}xQ^{-u}, \qquad \mathsf{E}_{uj} = b_u^j, \qquad G_{uu} = q^{du}.$$

The matrix E evaluates a polynomial at the N distinct points b_u. With T written in the monomial basis,

$$\mathsf{E}T\mathsf{E}^{-1} = -G^{-1}\Omega_s^{\mathsf{T}}G. \tag{4.3}$$

Consequently the full original pair matrix satisfies $\chi_N(\Omega) = 0$, including its nilpotent parts.

Proof. At $z = b_u$, the two physical clocks are

$$-q^{-1}x/b_u = Q^u, \qquad -q^{-1}b_u/y = Q^{s-u}.$$

The coefficients in (4.2) are therefore $-D$, $q^{-1}A_+$, and qA_-. Since $q^d = -q^{-1}$, the positive coefficient is $-q^dA_+$, and the negative one is $-q^{-d}A_-$. Evaluation at $Q^{-1}b_u$ increments the first index, while evaluation at Qb_u decrements it. The endpoint zeros in (4.1) kill every attempted wrap. Thus the coefficient from row u to column r is $-q^{d(r-u)}(\Omega_s)_{ru}$, proving (4.3). The transpose occurs because evaluation acts on functions of the input index, whereas the physical matrix sends input columns to output rows.

The matrix of $-T$ is triangular, with diagonal $t_j + t_j^{-1}$, where $t_j = qQ^j$. Its characteristic polynomial is

$$\prod_{j=0}^{N-1}(X - t_j - t_j^{-1}) = 2T_N(X/2) + 2.$$

To check the identity with multiplicities, substitute $X = w + w^{-1}$ and multiply by w^N. The left side becomes

$$\prod_{j=0}^{N-1}(w - t_j)(w - t_j^{-1}) = (w^N + 1)^2,$$

because both lists are exactly the roots of $t^N = -1$. The right side gives the same Laurent polynomial. Substitution $X = w + w^{-1}$ is injective on polynomials, so the identity follows. Cayley–Hamilton and (4.3) give $\chi_N(\Omega_s) = 0$. The N residue spaces form a direct sum of the full pair space, proving the assertion for Ω. This proves a matrix identity, not only an identity on its eigenvalues. □

The same transform gives the complete local polynomial adapter. For any polynomial f, write

$$f(-T)z^j = \sum_{\alpha,\beta\geq 0,\ \alpha+\beta\leq j} M_f(j;\alpha,\beta)x^\alpha y^\beta z^{j-\alpha-\beta}$$

and define

$$H_f(x,b,c,y) = \sum_{j=0}^{d}\sum_{\alpha+\beta\leq j} M_f(j;\alpha,\beta)x^\alpha b^{j-\alpha-\beta}c^{d-j}y^\beta. \tag{4.4}$$

Its exponents are nonnegative and its total degree is d, regardless of the degree of f. Fourier inversion is particularly simple:

$$(\mathsf{E}^{-1})_{jr} = \frac{1}{N}b_r^{-j}.$$

For a polynomial f, (4.3) gives

$$\mathsf{E}f(-T)\mathsf{E}^{-1} = G^{-1}f(\Omega_s)^{\mathsf{T}}G.$$

Use the displayed inverse of E. For $b = b_u, c = b_r$, the result is

$$(f(\Omega_s))_{ru} = \frac{1}{N}q^{-d(r-u)}c^{-d}H_f(x,b,c,y). \tag{4.5}$$

This proves the factor $1/N$, the direction of the shift and the full polynomial identity directly from the original pair matrices. If the input and output have different integer totals, the left

side is zero: the modular block is the direct sum of its actual-total blocks. Hence the right side also vanishes at such a grid point.

At even levels $q^{d(p+N)} = -q^{dp}$. The exponent $r - u$ in (4.5) is therefore the canonical integer displacement, not an arbitrarily chosen residue. The endpoint calculation above is what permits the Fourier transform; it does not assume that this phase is periodic. With the coherent face-logarithm lifts, $c = Q^{-(r-u)}b$ converts the prefactor to $N^{-1}(bc)^{-d/2}$, with these coherent choices of logarithms. This rewriting alone does not identify a closed diagram contraction.

This argument holds at every fixed root with $N \geq 2$, including composite levels. It proves the annihilator and the adapter for polynomials of the actual Casimir. The polynomial representing a crossing power and its repeated-node derivatives are established in Section 5. The comparison with a specified knot invariant requires its marked diagram adapter. No derivative with respect to a generic deformation of q is taken here.

5 Crossing values and their repeated-root normals

The Gaussian multiplier was defined by its germs, so the norm theorem is independent of any matrix convention. We now identify it with powers of an explicitly specified finite crossing. This also explains why the first derivatives in the functional are required. The global comparison of a marked diagram with its shared polynomial remains a separate step.

Let $z \in \mathbb{C}^*$ first be non-root-of-unity, write $m = N - 1$, and set $\rho_z = z - z^{-1}$. On the basis $v_0, \dots, v_m$ define

$$\begin{aligned} Hv_j &= (m-2j)v_j, & Kv_j &= z^{m-2j}v_j, \\ Ev_j &= [j]_z[m-j+1]_z v_{j-1}, & Fv_j &= v_{j+1}, \end{aligned} \qquad [j]_z = \frac{z^j - z^{-j}}{z - z^{-1}}.$$

As before, endpoint vectors outside the range are zero. Define $[k]_z! = \prod_{r=1}^{k}[r]_z$, with $[0]_z! = 1$, and put

$$\mathscr{R}_z = z^{H\otimes H/2}\sum_{k=0}^{m}\alpha_k E^k \otimes F^k, \qquad \alpha_k = \frac{z^{k(k-1)/2}\rho_z^k}{[k]_z!}, \qquad B_z = \mathsf{P}\mathscr{R}_z,$$

where $\mathsf{P}(v \otimes w) = w \otimes v$. Half-integral powers are read on a fixed local square-root branch; at $z = q$ this is the branch $q^a = \exp(\pi i a/N)$. The pair operators are

$$E_{12} = E \otimes K + I \otimes E, \quad F_{12} = F \otimes I + K^{-1} \otimes F, \quad K_{12} = K \otimes K,$$

and their Casimir is $\Omega_z = \rho_z^2 F_{12}E_{12} + zK_{12} + z^{-1}K_{12}^{-1}$. The specialization of this matrix in the balanced basis is the Ω of Section 4.

Proposition 5.1 (The finite crossing polynomial)**.** *At $z = q$, the displayed finite crossing satisfies*

$$B_q = q^{d^2/2}r_N(\Omega), \qquad B_q^n = \widetilde{f}_n(\Omega) \quad (n \in \mathbb{Z}),$$

where r_N and $\widetilde{f}_n$ are the polynomials defined in (3.9). These are full matrix identities.

Proof. We first verify the intertwining that will be used to find the scalar on each generic summand. The displayed matrices satisfy $KEK^{-1} = z^2E$, $KFK^{-1} = z^{-2}F$, and $[E, F] =$

$(K - K^{-1})/\rho_z$. Induction gives

$$[E, F^k] = \frac{[k]_z}{\rho_z} F^{k-1}(z^{-(k-1)}K - z^{k-1}K^{-1}),$$
$$[E^k, F] = \frac{[k]_z}{\rho_z} E^{k-1}(z^{k-1}K - z^{-(k-1)}K^{-1}).$$

Write $A = \sum_k \alpha_k E^k \otimes F^k$ and $D = z^{H\otimes H/2}$. Moving D through the pair generators reduces the two required intertwining identities to

$$A(E \otimes K + I \otimes E) = (I \otimes E + E \otimes K^{-1})A,$$
$$A(F \otimes I + K^{-1} \otimes F) = (K \otimes F + F \otimes I)A.$$

For $0 \le k < m$, the coefficient in the first difference multiplying $E^{k+1} \otimes F^k(K - z^{2k}K^{-1})$ is $\alpha_k - \alpha_{k+1}[k+1]_z z^{-k}/\rho_z$. For the second, the coefficient multiplying $E^k(z^k K - z^{-k}K^{-1}) \otimes F^{k+1}$ is $\alpha_{k+1}[k+1]_z/\rho_z - \alpha_k z^k$. Both vanish by the displayed formula for α_k; the terminal terms vanish because $E^N = F^N = 0$. Intertwining with K_{12} follows from the weights. Hence B_z commutes with E_{12}, F_{12}, K_{12}. The same commutation relations for these pair generators give $[\Omega_z, E_{12}] = [\Omega_z, F_{12}] = [\Omega_z, K_{12}] = 0$ by substitution into the displayed formula for Ω_z.

For $0 \le k \le m$, seek a highest vector

$$h_k = \sum_{a=0}^{k} b_a v_a \otimes v_{k-a}, \qquad b_0 = 1.$$

The equation $E_{12}h_k = 0$ gives

$$\frac{b_{a+1}}{b_a} = -\frac{[k-a]_z[m-k+a+1]_z}{[a+1]_z[m-a]_z\, z^{m-2k+2a+2}}, \qquad b_k = (-1)^k z^{-k(m-k+1)}.$$

The last formula follows by multiplying the ratios: the quantum-integer factors cancel in reverse order. The highest weight is $2m-2k$. For a highest vector of weight λ, direct induction gives

$$E_{12}F_{12}^r h = [r]_z[\lambda - r + 1]_z F_{12}^{r-1} h.$$

It follows that h_k generates a module of dimension $2m - 2k + 1$: none of these vectors can vanish before that length. The next vector, if nonzero, would be a highest vector of negative integral weight. For a negative highest weight λ, both $[r]_z$ and $[\lambda - r + 1]_z$ are nonzero for every $r \ge 1$, since z is not a root of unity. Induction in the same identity would force every lowering to be nonzero, contrary to nilpotence of F_{12}. The module is irreducible because any nonzero invariant subspace contains a weight vector: the distinct eigenvalues of K_{12} allow its weight projections to be written as polynomials in K_{12}. That vector can be raised to h_k and then lowered through the whole module.

On this module the Casimir scalar is

$$\eta_k(z) = z^{2N-1-2k} + z^{-2N+1+2k}.$$

For generic z these scalars are distinct, and the displayed dimensions sum to N^2. Thus the modules form a direct sum of the full pair space. Since B_z commutes with the generators, it is

scalar on each one: it preserves the one-dimensional highest-weight line, and commutation with F_{12} fixes the same scalar on every lowering. To compute its scalar on h_k, examine the coefficient of $v_0 \otimes v_k$ in $\mathsf{P}\mathscr{R}_z h_k$. Only the zero-th term of the sum defining $\mathscr{R}_z$, acting on $v_k \otimes v_0$, can contribute. Therefore the scalar is

$$\lambda_k(z) = (-1)^k z^{m^2/2 - k(2N-1-k)}.$$

Ordinary interpolation at the distinct generic scalars supplies a polynomial p_z of degree less than N such that $B_z = p_z(\Omega_z)$.

It remains to specialize this identity without discarding the normal part. At $z = q$ the nodes k and $l = N - 1 - k$ collide, and their crossing values agree. Put $h = 2k + 1 < N$ and $A_h = q^h - q^{-h} \neq 0$. Differentiating with respect to $\log z$ gives

$$\begin{aligned} \partial_{\log z}(\eta_k - \eta_l)|_q &= -2N A_h, \\ \partial_{\log z}(\lambda_k - \lambda_l)|_q &= \lambda_k(q) N (N - h). \end{aligned}$$

Consequently the limiting divided difference is

$$\lim_{z \to q} \frac{\lambda_k(z) - \lambda_l(z)}{\eta_k(z) - \eta_l(z)} = -\frac{\lambda_k(q)(N-h)}{2(q^h - q^{-h})}.$$

Replace each colliding pair of interpolation rows by its first row and its divided-difference row. The coefficient matrix tends to the confluent Vandermonde matrix for the double roots c_a and the odd simple root when present. It is invertible: a polynomial of degree less than N in its kernel has all the prescribed zero jets and is therefore divisible by χ_N. Thus the coefficients of p_z are regular at q, and their limits are the specified Hermite data. The value is $\lambda_k(q) = q^{d^2/2}(-1)^k q^{k(k+1)}$.

Every entry of B_z is regular there as well, since $[j]_q! \neq 0$ for $j < N$. Specializing the matrix identity proves $B_q = q^{d^2/2} r_N(\Omega)$ with its nilpotent part included. The nodal values of r_N are nonzero, so its class modulo χ_N is invertible. An inverse is obtained by prescribing the jets $r_N(c)^{-1}$ and $-r_N'(c)/r_N(c)^2$ at each double root, and the inverse value at the odd simple root. The product has the same jets as one, so is one modulo χ_N. Now (3.9), together with $\chi_N(\Omega) = 0$, proves the assertion for every signed power. □

The specialization argument uses a generic parameter to establish a finite matrix identity, not to define a transverse derivative of the knot invariant. Its derivative is used only to recover the coefficient of the repeated spectral root. No generic projector or generic direct sum is specialized at the root.

6 An elementary exact derivative inequality

For every complex trigonometric polynomial P of degree at most $d \geq 1$, set $\theta_j = (2j-1)\pi/(2d)$, $1 \leq j \leq 2d$. The classical Riesz interpolation formula [2] is

$$P'(x) = \sum_{j=1}^{2d} \frac{(-1)^{j+1}}{4d \sin^2(\theta_j/2)} P(x + \theta_j). \tag{6.1}$$

For completeness, invert the finite Fourier matrix on modes $-d+1,\dots,d$. If $z=e^{-i\theta_j}$, so $z^{2d}=-1$, differentiating the finite geometric sum gives its weight

$$\frac{1}{2d}\sum_{\ell=-d+1}^{d} i\ell e^{-i\ell\theta_j}=\frac{iz^{-d+1}}{d(1-z)^2}=\frac{(-1)^{j+1}}{4d\sin^2(\theta_j/2)}. \tag{6.2}$$

The extra mode $-d$ has samples opposite to those of d, and its derivative at zero is opposite too. Thus (6.1) holds on all modes $-d,\dots,d$; translation gives arbitrary x.

The elementary sine product $\prod_{j=0}^{2d-1}\sin(x+j\pi/(2d))=C_d\sin(2dx)$, with $C_d\neq 0$, yields after two logarithmic derivatives and substitution $x=\pi/(4d)$

$$\sum_{j=1}^{2d}\frac{1}{4d\sin^2(\theta_j/2)}=d. \tag{6.3}$$

Taking absolute values in (6.1) therefore proves

$$\|P'\|_{\mathbb{R}}\le d\|P\|_{\mathbb{R}}. \tag{6.4}$$

The argument works without real coefficients or positivity.

The same finite identity and triangle inequality hold when the coefficients lie in any finite-dimensional normed space. Thus (6.4), and consequently (1.1)–(1.4), hold for such vector-valued polynomials without a dimension factor. The intended application here remains the original shared scalar coefficient of Section 11; no change of the source norm is required.

7 Full and vanishing-node bounds

Since H has degree at most d, $P(\phi)=H(2\cos\phi)$ has trigonometric degree at most d, and its supremum is exactly $\|H\|_{[-2,2]}$. Equations (3.7), (3.10), and (6.4) give

$$|\mathcal{L}_N^n H|\le\frac{N}{N^2}\left(1+\frac{|n|N}{2}+d\right)\|H\|=\left(1+\frac{|n|}{2}\right)\|H\|, \tag{7.1}$$

because $d+1=N$. This proves (1.1).

If H vanishes at every node, the first two terms of (3.10) vanish exactly. Only the derivative of the original numerator remains. Applying (6.4) to that term gives (1.2), uniformly in every signed n, including zero.

Node vanishing does not make the functional zero. For any double root c_a, the polynomial

$$H_a(z)=\frac{\chi_N(z)}{z-c_a}$$

has degree d and vanishes at every spectral node, including the odd simple node. At c_a its derivative is $h(c_a)$, and every other term in (3.3) is zero. Hence

$$\mathcal{L}_N^n H_a=f_n(c_a),\qquad |\mathcal{L}_N^n H_a|=1. \tag{7.2}$$

The improvement in the norm is therefore a bound on the surviving normal contribution, not a claim that all values determine the answer.

8 Shared polynomials and the unsplit complementary source

The other finite functionals preserve each partial node-vanishing condition. Applying (1.1) and (1.2) successively proves (1.3). More explicitly, after the other variables are fixed on their real intervals, each application leaves a polynomial of the same degree bound in the remaining variables. A vanishing condition in a different coordinate is preserved by linearity. The same finite Riesz identity controls derivatives with values in the remaining finite-dimensional polynomial space, with its real-cube supremum norm; there is no dimension factor.

For clarity, (1.4) needs only joint grid vanishing, not vanishing on every node hyperplane. In each coordinate (3.10) is a sum of an evaluation term and a term differentiating P. Before summing its N grid entries, the two coefficient bounds are $1 + |n_e|N/2$ and 1, respectively. For a subset A of differentiated variables, iterating (6.4) gives

$$\|\partial_A \mathcal{D}(2\cos\boldsymbol{\phi})\| \le d^{|A|}\|\mathcal{D}\|_{[-2,2]^r}. \tag{8.1}$$

The term $A = \varnothing$ vanishes pointwise at the tensor grid. Every other term is bounded by $\prod_{e\in A}(d/N)\prod_{e\notin A}(1/N + |n_e|/2)\|\mathcal{D}\|$. Adding the nonempty subsets proves (1.4).

The joint condition arises naturally from a product of local factors. Suppose P_e and C_e have identical values at every spectral node in their e-th variable, as polynomials in all shared auxiliary variables. Then $\prod_e P_e - \prod_e C_e$ vanishes on the full spectral product grid. Any fixed linear extraction of the auxiliary variables preserves this identity. Thus (1.4) applies after that shared extraction, provided the degree in each spectral variable is at most d. This argument does not require vanishing on each node hyperplane.

For example, if $N = 3$ then $\chi_3(z) = (z-1)^2(z+2)$. The polynomial $(z-1)(z+2)$ vanishes at both distinct nodes but has nonzero first derivative at the double root. In several variables a sum of products with different such factors can vanish on the product grid without sharing a factor in any single coordinate. The subtraction of the all-evaluation term in (1.4) remains valid in that case.

9 Exact evaluation of a marked diagram

The local polynomial now has a global interpretation. A variable belongs to a face of the diagram, so all its occurrences at different boxes must be identified before a coefficient is taken. We prove that this shared coefficient is the reduced tensor evaluation, including its single marked cap. No positivity or limiting estimate enters the proof.

9.1 The ordered diagram and the coefficient theorem

Throughout this section $N \ge 3$, $d = N - 1$, $q = e^{\pi i/N}$, $Q = q^2$, and $x_0 = q^d$. Start with a finite connected triangulation of the oriented sphere and a perfect matching of its cubic dual. We allow a triangular cellulation, but require that the two triangles at each matching edge form an embedded quadrilateral disk with four distinct vertices. There are $2R$ triangles, R matching edges and $R + 2$ vertices. The vertices will be called *white faces*, because they index the complementary regions in the diagram below.

Contract a small disk around each matched dual edge and its endpoints. The disks are disjoint, and the resulting embedded graph Γ is connected and four-valent, with R vertices

and $2R$ edges. Choose an ordered parametrization of every disk. Its four surrounding white faces have cyclic order

$$(x, b, y, c);$$

the matching diagonal is xy. The coherently oriented triangles can be written (x, b, y) and (y, c, x). The polynomial of this box is $H_f(x, b, c, y)$: b is its input middle face and c its output middle face.

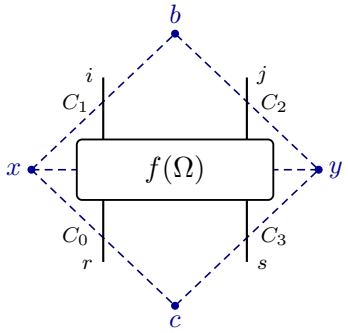
Here is the terminal convention. Write the four terminals as C_0, C_1, C_2, C_3 in cyclic order, with input terminals C_1, C_2 and output terminals C_0, C_3. The matching halves group (C_1, C_2) and (C_3, C_0); the untwisted physical strands pair (C_0, C_1) and (C_2, C_3). These are different pairings. When side indices $0, 1, 2, 3$ are used for a quadrilateral, as in an annular braid drawing, they denote

$$(C_1, C_2, C_3, C_0),$$

respectively. In those indices the physical pairs are $(0, 3)$ and $(1, 2)$, and the matching halves are $(0, 1)$ and $(2, 3)$.

Draw the two untwisted strands in each disk and join them along the uncontracted dual edges. The outside arcs are disjoint. They can be made Morse with respect to height, keeping the disk parametrizations fixed, so that each maximum or minimum is an ordinary cap or cup. Place the specified two-strand operator in each disk. A crossing power uses the checked crossing of Proposition 5.1; an arbitrary polynomial in the Casimir is also allowed. The local frames are part of the data and are not reset when an arc is bent.

This construction is a literal planar tensor diagram. If crossing circles are desired, place one perpendicular circle around the two strands in each disjoint three-ball about a disk. Its crossing disk meets precisely those two strands. Thus the corresponding augmented link is embedded in S^3; no geometric realization assumption is needed for the finite evaluation. An arbitrary polynomial coupon itself need not be a braid or define a knot.



$i + j = r + s$



$$i - \frac{d}{2} = \lambda_x - \lambda_b$$

$$L_f = \exp(2\pi \mathrm{i} \lambda_f / N)$$

$$Q^i = q^d L_x / L_b$$

Centered head and face potentials

Figure 1: The ordered box has cyclic face order (x, b, y, c) and polynomial $H_f(x, b, c, y)$. Its two inputs are C_1, C_2, and its outputs are C_0, C_3. The matching diagonal xy is not a physical strand. On a supported state the displayed head satisfies $i - d/2 = \lambda_x - \lambda_b$; this fixes the sign of the face potential. The alternative side numbering $0, 1, 2, 3$ refers to C_1, C_2, C_3, C_0.

A main edge means one of the edges outside the box disks. Choose one as a marked edge and call its adjacent faces a, b. They are distinct. Indeed a connected four-valent graph has no bridge: if removing a single edge separated a set of vertices, the sum of their degrees would be

twice the number of internal edges plus one, contrary to its even parity. An edge incident to only one face of a spherical graph is a bridge.

The matrices defining the turns and mark are given below. Let $A_{\mathrm{raw}}(f_1, \dots, f_R)$ be their contraction, with exactly one marked cap and with the full operators $f_e(\Omega)$ in the recorded boxes. Each f_e is a finite polynomial; its coefficients may depend on N and on other parameters. In particular it may already include a signed crossing power through Proposition 5.1. Set

$$\mathcal{P}(L) = \prod_{e=1}^{R} H_{f_e}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e}).$$

Every local factor is homogeneous of degree d, so $\deg \mathcal{P} = dR$.

Theorem 9.1 (The exact marked face coefficient)**.** *For the stated ordered diagram and the full polynomial coupons,*

$$A_{\mathrm{raw}}(f_1, \dots, f_R) = \left[\prod_{f \neq a,b} L_f^d \right] \mathcal{P}(L)|_{L_a = L_b = 0}\,. \tag{9.1}$$

This is an identity at each level, with no omitted power of N. The reduced scalar is independent of the chosen main edge when the cut and its duality frames are transported as below. The theorem does not assert this independence for separate projections of a local polynomial onto selected monomials.

We prove the theorem in several finite steps. The first concerns the reduced cut; the remaining steps turn its state sum into the displayed coefficient.

9.2 Duality, turns and the unrestricted mark

For a generic nonzero parameter z which is not a root of unity, let V_L have basis $v_0, \dots, v_L$ and operators

$$Kv_i = z^{L-2i} v_i, \qquad Fv_i = v_{i+1}, \qquad Ev_i = [i]_z [L - i + 1]_z v_{i-1}.$$

Endpoint vectors are zero. The coproducts are those used in Section 5, and the antipodes are

$$S(E) = -EK^{-1}, \qquad S(F) = -KF, \qquad S(K) = K^{-1}.$$

The dual action is obtained by transposing these antipode matrices. Put

$$J_z v_i = c_i(z) v^*_{L-i}, \qquad c_i(z) = (-1)^i z^{i(i-L-1)}. \tag{9.2}$$

This is an intertwining map from V_L to its dual. For F, the required ratio is

$$c_{i+1}/c_i = -z^{-L+2i};$$

the K equation reverses the weights, and the E equation then uses the same ratio and the displayed quantum integers. These checks also show that J_z and its inverse are regular near our root when $L = d$.

The integer-weight operator H satisfies $K = z^H$. Since E and F change weights by two, $(-1)^H$ is central and group-like. Consequently

$$\widetilde{g} = (-1)^H K$$

implements S^2 on every tensor product and dual used here. In the given self-dual frame it is exactly the double-dual map:

$$J_z^{-t} J_z = \widetilde{g}, \qquad \frac{c_i(z)}{c_{L-i}(z)} = (-1)^L z^{L-2i}.$$

At $z = q$, every integral weight satisfies $q^{-NH} = (-1)^H$. Thus the generic pivot specializes to the physical root pivot

$$g = K^{1-N}. \tag{9.3}$$

This specialization uses a regular generic pivot; it does not assert that K^{1-N} implements S^2 for a generic z.

Take $L = d$ and pass to the balanced basis $e_i = v_i/(Q;Q)_i$ at the root. The product identity

$$(Q;Q)_i (Q;Q)_{d-i} = N(-1)^i Q^{i(i+1)/2} \tag{9.4}$$

follows from $(Q;Q)_d = N$ and $1 - Q^{-j} = -Q^{-j}(1 - Q^j)$. Since $c_i(q) = q^{i^2}$, the factorials in J_q therefore cancel exactly. If the first tensor factor below is a dual factor, denote it by a superscript $\vee$. The four ordered turns are

$$\begin{aligned} \mathrm{ev}_L(e_a^\vee \otimes e_b) &= N^{-1} x_0^a \, \mathbf{1}_{a+b=d}, \qquad \mathrm{coev}_L(1) = \sum_a N x_0^{-a} e_{d-a} \otimes e_a^\vee, \\ \mathrm{ev}_R(e_b \otimes e_a^\vee) &= N^{-1} x_0^a g_b \, \mathbf{1}_{a+b=d}, \quad \mathrm{coev}_R(1) = \sum_a N x_0^{-a} g_{d-a}^{-1} e_a^\vee \otimes e_{d-a}, \end{aligned}$$

where

$$g_j = q^{-d^2-2j}.$$

Contraction of a turn with the opposite turn has a single matching index; its x_0 and g factors cancel. This proves all four snake identities. Closing an endomorphism by the corresponding ordinary turns gives the pivotal trace $\mathrm{Tr}(gA)$.

In literal left-to-right physical head order (i, j), both cups and both caps reduce to

$$p_{ij} = N q^{-d^2+di} \mathbf{1}_{i+j=d}, \qquad \ell_{ij} = N^{-1} q^{di} \mathbf{1}_{i+j=d}. \tag{9.5}$$

For example the right-cap ratio to the stated cap is $q^{-2Ni} = 1$, and the right-cup ratio is $q^{2N(d-i)} = 1$. These are equalities of the actual turn matrices, not independent rescalings of the local frames.

Define the rank-one projector $e = \psi\phi$ by $\psi_i = 1$, $\phi_i = 1/N$, so $e^2 = e$ and $\mathrm{Tr}\, e = 1$. At a right cut insert $g^{-1}e$ on the primal input. For every matrix C,

$$\mathrm{Tr}(g\, g^{-1} e C) = \mathrm{Tr}(eC).$$

At a left cut use the corresponding defect ge. In physical left-to-right order the resulting marked rows are

$$m_R(i,j) = \frac{x_0^j}{N^2}, \qquad m_L(i,j) = \frac{x_0^i g_{d-i}}{N^2}. \tag{9.6}$$

For the right row, the only summand in $\mathrm{ev}_R(g^{-1}e\,e_i\otimes e_j^\vee)$ has primal index $d-j$, and its pivot times defect entry is $1/N$. This gives the first formula; the left formula follows by the same single-index contraction. Neither marked row has a support restriction.

Each complete marked cup-cap contraction is one. In a whole closed diagram with one marked cap, conservation forces the marked heads to satisfy $i+j=d$: each box preserves the true integer sum of its two indices, every cup supplies d, and every other cap consumes d. On this forced support both rows in (9.6) equal

$$m_{ij}=N^{-2}q^{dj}. \tag{9.7}$$

We shall use (9.7) only after this global conservation argument. Restricting the marked row in advance would be a different tensor.

9.3 Why the chosen reduced cut can be exposed

We next justify moving the marked edge to an exposed top cap while keeping every coupon disk and its ordered frame fixed. This is the only place where a closed spherical reading of the diagram is compared with a different planar reading.

For the generic pivot $\widetilde{g}$, the two traces agree on every intertwiner in the tensor-generated category:

$$\mathrm{Tr}(\widetilde{g}A)=\mathrm{Tr}(\widetilde{g}^{-1}A). \tag{9.8}$$

Here is a direct proof including multiplicity spaces. First take real $z>0$, $z\neq 1$. The involution

$$K^*=K,\qquad E^*=KF,\qquad F^*=EK^{-1}$$

is compatible with the stated coproducts. On V_L, choose orthogonal basis vectors with squared norms

$$n_0=1,\qquad n_i=[i]_z[L-i+1]_z\,z^{-L+2i}n_{i-1}>0.$$

These give precisely the displayed adjoint relations. Tensor products are unitary for the tensor-product norm; duals are unitarized by (9.2). An invariant subspace therefore has an invariant orthogonal complement.

Every irreducible summand is one of the V_L. Indeed choose a highest vector v of integral weight L. The identity

$$EF^rv=[r]_z[L-r+1]_zF^{r-1}v$$

shows that the first zero lowering occurs at $r=L+1$; a negative highest weight would have no zero lowering. Bounded weights exclude that possibility. The preceding nonzero lowerings have distinct weights and span the irreducible summand. An intertwiner consequently has blocks $I_{V_L}\otimes A_L$. On each such block both sides of (9.8) equal

$$(-1)^L[L+1]_z\,\mathrm{Tr}(A_L),$$

because the weights $L,L-2,\dots,-L$ are symmetric.

For a fixed diagram all matrix entries are rational in a fixed local square root of z. Polynomial coupon coefficients are held fixed while taking this generic lift. The identity on an interval of positive real z proves the rational identity, and all original matrices are regular at q. For the crossing this regularity was proved in Proposition 5.1; its inverse is regular as

well because its root value is invertible. Thus (9.8) specializes to the actual diagram. No decomposition or projection onto the generic irreducible summands is specialized.

Choose a point in face a and move it to the point at infinity along a path outside all coupon disks. A thin strip around that path gives a sphere isotopy which fixes the disks pointwise. The outside arcs may be crossed by the moving infinity, but no crossing or ribbon twist is introduced. Ordinary portions of the motion are tensor interchange and the four snake identities. When infinity passes an arc, the two planar readings close the same bent endomorphism by its two pivotal traces. Equality (9.8) gives the same generic closed scalar. Face a is now exterior.

The chosen edge borders that exterior face. Push a short subarc along a narrow strip within the face to a point above the remaining compact drawing. Its tip is a top cap; the strip introduces no crossing or extra component. The inside face is still b. Perturb the other arcs to make their turns Morse.

Leave the new cap open. After the actual cup identification, the remaining diagram is an endomorphism of V_d or its dual. Generically it is a scalar $a(z)I$: commuting with the distinct K weights makes it diagonal, and commuting with the nonzero adjacent F arrows makes its diagonal constant. The scalar is regular at q, since it is an entry of a regular open network. Closing it ordinarily gives

$$(-1)^d[N]_z\, a(z).$$

This factor is the same for every exposed cut, even on a different component, because every component has the same color. It is nonzero generically, so equal closed scalars have equal regular open scalars. Specialization gives their equality at q, without division by $[N]_q = 0$. The marked closure evaluates $\mathrm{Tr}(e\, aI) = a$. This proves the required independence of the reduced cut.

This argument does not commute the nonintertwining defect $g^{-1}e$ through the boxes. It first transports the cut with its true cups and duality, and then identifies its reduced scalar.

9.4 The affine face lattice and the true support

We describe the state space before evaluating the turn factors. For a vertical head separating a left face f from a right face g, use the convention

$$i - \frac{d}{2} = \lambda_f - \lambda_g, \qquad L_f = \exp(2\pi i \lambda_f / N). \tag{9.9}$$

At the first input of a box this reads $Q^i = q^d L_x/L_b = -q^{-1}L_x/L_b$, exactly the first clock in the local Fourier proof. At the other input the ratio is $q^d L_b/L_y$; the two output ratios use c in place of b. Reversing an arc replaces its head i by $d - i$, so its centered value changes sign.

A physical state on the unmarked diagram is a choice of canonical heads $0, \dots, d$. At a turn the heads sum to d; at every box the two input heads have the same integer sum as the two output heads. Centered heads thus define an antisymmetric flow on the edges of Γ, with zero divergence at each vertex. Equivalently, they give a closed cochain on the dual face graph.

For clarity, the required integral fact holds over every abelian group. A zero-divergence flow on a connected spherical graph is the difference of face potentials, unique after one potential is fixed. To prove it, integrate the dual cochain from the fixed face along dual edge paths. Two paths differ by a closed dual walk. Splitting repeated vertices reduces that walk to simple cycles; on the sphere each such cycle bounds a union of dual two-cells. Its cochain sum is

the sum of the divergences at the enclosed primal vertices, hence zero. This proves path independence. Connectedness gives uniqueness. Integration uses only addition and subtraction, so it introduces no finite index or determinant factor.

For odd N, the centered heads are integral. Fix $\lambda_a = 0$; their potentials are then integers. For even N, twice every centered head is odd. The corresponding parity cochain equals one on each main edge and zero on a matching diagonal: the boundary of each triangle has two main edges. It is closed, and the same spherical argument over $\mathbb{F}_2$ gives unique face parities t_f, with

$$t_a = 0, \qquad t_b = 1.$$

Equivalently, the dual quadrilateral at each box has four odd edges and opposite vertices of the same parity. The exact potentials have the form

$$\lambda_f = p_f + \frac{t_f}{2}, \qquad p_f \in \mathbb{Z}. \tag{9.10}$$

For odd N, use $t_f = 0$ throughout. These affine lattices are nonempty: the potential $t_f/2$ itself gives centered flow $(t_f - t_g)/2$, and its heads lie in the physical range. The difference of any two lifts is an integral face potential.

Reduce the p_f modulo N, with $p_a = 0$. There are exactly N^{R+1} face-grid points

$$L_a = 1, \qquad L_f = q^{t_f} Q^{p_f} \quad (f \neq a). \tag{9.11}$$

At every such point take the canonical head representing $d/2 + \lambda_f - \lambda_g$ modulo N. At a turn the congruence $i + j = d \pmod N$ forces the equality $i + j = d$: neither $d - N = -1$ nor $d + N = 2N - 1$ lies in $[0, 2d]$.

At a box the residue total need not determine the true integer total. This is the place where physical support is essential. The local Fourier proposition expresses the actual residue block as a direct sum of its true-total blocks. Hence if the canonical input and output totals differ, the matrix entry vanishes, and (4.5) implies

$$H_{f_e}(L_x, L_b, L_c, L_y) = 0 \tag{9.12}$$

at that grid point. The prefactor there is nonzero.

If no box violates the true total, the canonical heads form the closed centered flow just described. Integrating it gives unique exact potentials with $\lambda_a = 0$, and they reduce to the given grid point. Conversely every supported canonical state gives one grid point by this integration. Additional zeros of a coupon merely give zero weight on both sides. We have proved a bijection between supported states and supported grid points, and proved that all other grid points contribute zero.

The exact potentials of a supported state, rather than arbitrary representatives $0, \dots, N-1$ for the p_f, are used in intermediate half-powers. In particular at a box

$$r - u = \lambda_b - \lambda_c$$

is the canonical integer displacement, since b, c have the same parity. Thus

$$q^{-d(r-u)} c^{-d} = (bc)^{-d/2}, \qquad f_e(\Omega)_{r,s;u,v} = \frac{1}{N} \frac{H_{f_e}(x, b, c, y)}{(bc)^{d/2}}. \tag{9.13}$$

Here each fractional power means the exponential of the exact potential in (9.9). At even N one has $q^{d(p+N)} = -q^{dp}$; the proof has not asserted otherwise. An unsupported grid point needs no such lift, because its product already vanishes by (9.12).

9.5 Winding, the single cut factor and coefficient extraction

Let M_f be the number of input and output middle occurrences of face f at the boxes. Let T_f count outside turn occurrences positively and inside turn occurrences negatively, temporarily treating the marked cap as ordinary. Then

$$M_f - T_f = 2 \quad (f \neq a), \qquad M_a - T_a = -2. \tag{9.14}$$

Indeed traverse each face boundary with the face on its left. Resolve repeated boundary passages by a narrow regular neighborhood. In the recorded local box resolution each middle passage reverses the vertical tangent and contributes π; the two outer passages have net turn zero. An inside cup or cap contributes π, and its outside passage contributes $-\pi$. The intervening monotone arcs have zero net contribution after deformation to vertical tangents. This also follows by counting signed passages through a horizontal tangent direction. The bounded-face boundary has winding $+1$, and the exterior boundary winding -1, proving (9.14). Every repeated occurrence of a face is counted, and no restriction on the relative heights of different boxes was used.

At a cup or cap with outside face o and inside face z, the first head satisfies

$$q^{di} = q^{d^2/2}(L_o/L_z)^{d/2}.$$

Thus an ordinary cup contributes $Nq^{-d^2/2}$ times this face character and a cap contributes $N^{-1}q^{d^2/2}$ times the same character. There are equally many cups and caps, so their constant phases and powers of N cancel before one cap is marked. At that exposed cap, (9.7) gives

$$\frac{m_{ij}}{\ell_{ij}} = N^{-1}q^{d(j-i)} = N^{-1}(L_b/L_a)^d. \tag{9.15}$$

Equations (9.13) and (9.14) therefore turn the whole literal marked summand into

$$N^{-R-1}\,\frac{\mathcal{P}(L)}{\prod_{f\neq a,b} L_f^d}. \tag{9.16}$$

Equivalently, with k cups, $k-1$ ordinary caps and one marked cap, the turn count is $N^k N^{-(k-1)}N^{-2} = N^{-1}$; each of the R local Fourier inversions supplies its separate N^{-1}. This is the single reduced-cut normalization.

All fractional face powers have disappeared from (9.16). Changing any exact potential by a multiple of N leaves this whole expression unchanged. By the supported-state argument it can be summed over the entire grid (9.11), not just over a chosen lift.

It remains to calculate this normalized grid average. A monomial of $\mathcal{P}$ has nonnegative face exponents n_f with total dR. Character orthogonality on each of the $R+1$ independent N-point grids gives the conditions

$$n_b \equiv 0 \pmod N, \qquad n_f \equiv d \pmod N \quad (f \neq a, b).$$

There are R ordinary faces, each with exponent at least d. Their minimum already exhausts the total degree dR. Consequently the only surviving exponents are

$$n_a = n_b = 0, \qquad n_f = d \quad (f \neq a, b). \tag{9.17}$$

At this exponent the even-coset character is exactly one. The N^{R+1} grids cancel the denominator in (9.16). This proves Theorem 9.1, with both parities and the entire normal data of every coupon retained.

9.6 Insertion of signed powers after the shared extraction

The preceding theorem is an equality for full polynomial operators. It may therefore be combined with resolvent reproduction before any estimate is taken. Define

$$S_N(\boldsymbol{\sigma}) = \left[\prod_{f\neq a,b} L_f^d\right] \prod_{e=1}^{R} H_{V_N(\sigma_e;\Omega)}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e})\Big|_{L_a=L_b=0}. \tag{9.18}$$

Each spectral degree is at most d. The extraction and multiplication in (9.18) take place in the same face variables as in the original diagram.

Corollary 9.2 (Exact filling of the marked source)**.** *For any fixed signed integer powers $n_1, \dots, n_R$, the raw marked evaluation with $B_q^{n_e}$ in box e is*

$$A_{\mathrm{raw}}(\boldsymbol{n}) = \left(\bigotimes_{e=1}^{R} \mathcal{L}_N^{n_e}\right) S_N(\boldsymbol{\sigma}). \tag{9.19}$$

If C_N is any other polynomial with the same separate spectral degree bound, then

$$\left|A_{\mathrm{raw}}(\boldsymbol{n}) - \left(\bigotimes_{e} \mathcal{L}_N^{n_e}\right) C_N\right| \le \prod_e (1+|n_e|/2)\, \|S_N - C_N\|_{[-2,2]^R}.$$

The joint-grid improvement of Theorem 1.1 applies when its vanishing hypothesis is satisfied.

Proof. Proposition 2.3 gives $\mathcal{L}_N^{n_e} V_N(\sigma_e;\Omega) = \widetilde{f}_{n_e}(\Omega)$ in each individual box. Proposition 5.1 identifies the latter with $B_q^{n_e}$. The ordered tensor contraction is multilinear in its local matrices, so these identities may be applied separately at every box and then Theorem 9.1 gives (9.19). No commutation between boxes is used. Subtract the two full shared polynomials before applying Theorem 1.1; this proves the norm assertion. □

The exact coefficient identity and the last corollary contain no asymptotic remainder hypothesis. A strict bound for the particular polynomial $S_N - C_N$, and any nonzero limiting contribution assigned to C_N, remain separate analytic questions.

10 Comparison with the reduced Kashaev invariant

We specify the invariant convention rather than leaving a phase or a change of basis implicit. This comparison is finite and applies at every $N \ge 2$. The global coefficient theorem above is used only at $N \ge 3$.

Let P exchange tensor factors. In this section set $R = \mathscr{R}_q$ and $B = B_q = \mathsf{P}R$, using the crossing defined in Section 5. Set

$$p_j = \prod_{r=1}^{j}(1-Q^r), \qquad \bar{p}_j = \prod_{r=1}^{j}(1-Q^{-r}), \qquad p_0 = \bar{p}_0 = 1.$$

For an integer a, let $[a]_N$ be its representative in $\{0,\dots,d\}$. The enhanced Kashaev crossing in the column-vector convention is

$$\begin{aligned}(B_K)_{ce,ab} &= \frac{NQ^{1+c-b+(a-e)(c-b)}\mathbf{1}_{u+t<N}\mathbf{1}_{v+h<N}}{p_u\bar{p}_v p_t \bar{p}_h},\\ (u,v,t,h) &= ([b-a-1]_N, [a-c]_N, [c-e]_N, [e-b]_N).\end{aligned} \tag{10.1}$$

Its enhancement is

$$\mu_K = -qS, \qquad \alpha_K = -q, \qquad \beta_K = 1, \qquad Sv_j = v_{[j+1]_N}.$$

These are the enhanced crossing and normalization of Murakami–Murakami [4, §§1,4], with their quantum parameter equal to Q. A negative braid generator uses the actual inverse of (10.1). The reduced value is obtained by cutting one strand, taking the μ_K-weighted traces on the others, and multiplying by α_K^{-w}, where w is the writhe of the braid word. Its unknot value is one. We use the standard enhanced ribbon invariant defined by these data; the calculation below proves the normalization comparison rather than invoking its comparison theorem.

Theorem 10.1 (Exact normalization of the marked coefficient)**.** *In the enhanced braid conventions just specified, a link L with c components satisfies*

$$F_N^g(L) = (-1)^{d(c-1)} K_N(L), \qquad g = K^{1-N}, \tag{10.2}$$

where F_N^g is the reduced g-pivotal evaluation normalized by its positive curl. For a knot the sign is one.

For a diagram D of a knot K in Theorem 9.1, with its boxes filled by the specified signed crossing powers, the zero-framed reduced invariant is

$$K_N(K) = \zeta_{N,D} A_{\mathrm{raw}}(\boldsymbol{n}), \qquad |\zeta_{N,D}| = 1. \tag{10.3}$$

The character $\zeta_{N,D}$ consists of the recorded integer framing corrections. For a closed braid of writhe w in the same convention, it is θ^{-w}, where $\theta = q^{-d^2/2}$. Consequently the right side of (9.19), with this same character, evaluates the original invariant.

Proof. We first compare the matrices, keeping all factorial changes of basis. Define $R^o = (\mathsf{P}B_K)^{\mathsf{T}}$ and the Fourier matrix $\mathsf{F}_{ij} = Q^{ij}/\sqrt{N}$. Put

$$A = (\mathsf{F}^{-1} \otimes \mathsf{F}^{-1}) R^o (\mathsf{F} \otimes \mathsf{F}).$$

We claim that its only nonzero entries are

$$A_{a-k,b+k;\,a,b} = Q^{1+(a-k)(b+1)} \frac{p_a p_{b+k}}{p_{a-k} p_b p_k}, \qquad 0 \le k \le \min(a, d-b). \tag{10.4}$$

In particular the conservation is equality of integer sums.

Here is the finite calculation. For a nonzero entry of R^o introduce the residues

$$u = [j-i-1]_N, \quad v = [i-l]_N, \quad t = [l-k]_N, \quad h = [k-j]_N.$$

The two indicators say $u + t < N$ and $v + h < N$; their total is congruent to -1 modulo N, so $u + v + t + h = d$. Conversely this composition and one free index j determine the original entry:

$$i = j - 1 - u, \quad k = j + h, \quad l = j + h + t \pmod N, \qquad (R^o)_{ij,kl} = \frac{N Q^{1-(t+h)(u+h)}}{p_u \bar{p}_v p_t \bar{p}_h}.$$

Summing the free index in the Fourier transform gives the modular conservation delta. If

$$S_r(X) = \sum_{j=0}^{r} \frac{X^j}{p_j \bar{p}_{r-j}},$$

the remaining sum is

$$A_{c,e;\,a,b} = \mathbf{1}_{a+b\equiv c+e \pmod N} \sum_{z=0}^{d} Q^{1+c-z^2+z(a+b)} S_{d-z}(Q^{c-z}) S_z(Q^{z-a}). \tag{10.5}$$

Use $\bar{p}_r = (-1)^r Q^{-r(r+1)/2} p_r$. Comparing coefficients in $(1-Q^r)S_r(X) = (X-Q^r)S_{r-1}(X)$ proves

$$S_r(X) = \frac{\prod_{j=1}^{r}(X-Q^j)}{p_r}.$$

The two factors in (10.5) now force $c \le z \le a$. With $k = a-c$, the surviving sum is

$$Q^{1+c(b+1)} \frac{p_a}{p_c} \sum_{j=0}^{k} \frac{(-1)^j Q^{j(j-1)/2} Q^{(b+1)j}}{p_j p_{k-j}} = Q^{1+c(b+1)} \frac{p_a}{p_c} \frac{(Q^{b+1};Q)_k}{p_k}.$$

Here $(a;Q)_k = \prod_{r=0}^{k-1}(1-aQ^r)$, with empty product one. The last equality is the same finite product identity after reversing its coefficients. If $b+k \ge N$, the numerator contains $1-Q^N$ and is zero. Otherwise $(Q^{b+1};Q)_k = p_{b+k}/p_b$. This proves (10.4) with its endpoint support.

Define diagonal matrices and a scalar by

$$G_j = \frac{p_j}{(1-Q)^j} = q^{j(j-1)/2}[j]_q!, \qquad D_j = (-q)^j, \qquad \gamma = Qq^{-d^2/2}.$$

The coefficient of $E^k \otimes F^k$ in the defining finite R has the factorial ratio $([a]_q!/[a-k]_q!)^2/[k]_q!$. Substitute $p_j = (1-Q)^j q^{j(j-1)/2}[j]_q!$, together with its Cartan factor. The resulting entry is exactly (10.4), giving

$$A = \gamma(DG \otimes G) R (G^{-1} \otimes G^{-1}D^{-1}). \tag{10.6}$$

For example, at the entry in (10.4) the multiplier is $\gamma D_{a-k} G_{a-k} G_{b+k}/(G_a G_b D_b)$. This fixes the positions of the factorial gauges; no estimate on their condition numbers has been used.

The Fourier calculation involves a transposed crossing. To compare it with the paper's $B = \mathsf{P}R$, regard the root self-duality as the antidiagonal matrix

$$Jv_j = c_j v_{d-j}, \qquad c_j = (-1)^j q^{j(j-N)}.$$

It obeys $JKJ^{-1} = K^{-1}$ and

$$(J \otimes J) R (J^{-1} \otimes J^{-1}) = R^{\mathsf{T}}. \tag{10.7}$$

For an entry shifted from (a,b) to $(a+k, b-k)$, reflection leaves the factorial ratio unchanged because $[N-j]_q = [j]_q$. Its J-ratio is

$$\frac{c_{d-a-k} c_{d-b+k}}{c_{d-a} c_{d-b}} = q^{2k(a-b)+2k^2}.$$

The reflected Cartan exponent is $H_a H_b/2 + k(H_a - H_b) - 2k^2$, with $H_a - H_b = 2(b-a)$; the extra exponents cancel. This proves (10.7) entry by entry, with the same endpoint zeros.

For a braid on s strands put

$$U_s = (\mathsf{F}^{-1})^{\otimes s} \bigotimes_{j=1}^{s} (G^{-1}D^{j-1}), \qquad W_s = U_s J^{\otimes s}.$$

Transposing (10.6) gives the adjacent crossing identity for $\mathsf{P}R^{\mathsf{T}}$. This crossing conserves the integer total, so it commutes with $D\otimes D$. The powers of D therefore combine into the displayed single U_s, valid at every adjacent position. For a signed braid word β of writhe w, one obtains

$$\rho_K(\beta) = \gamma^w W_s \rho_B(\beta) W_s^{-1}, \qquad W_s K^{\otimes s} W_s^{-1} = \mu_K^{\otimes s}. \tag{10.8}$$

For the enhancement identity, G, D commute with K^{-1}, $q^{-d} = -q$, and $\mathsf{F}^{-1}\,\mathrm{diag}(Q^j)\mathsf{F} = S$. Negative generators are handled by inverting the same adjacent-generator identity.

We give the cut-scalar argument explicitly. For an intertwiner T of a tensor prefix with one last factor, set $C = \mathrm{Tr}_{\mathrm{last}}((I\otimes K)T)$. The ordinary last trace of $[T, E_{\mathrm{prefix}}\otimes K + I\otimes E] = 0$ gives $[C, E_{\mathrm{prefix}}] = 0$. The K-weighted last trace of the corresponding F equation gives $[C, F_{\mathrm{prefix}}] = 0$: its other two terms cancel by cyclicity, using

$$T(K_{\mathrm{prefix}}^{-1}\otimes I) = (K_{\mathrm{prefix}}^{-1}\otimes K^{-1})T(I\otimes K).$$

Commutation with the prefix K is immediate. The argument can be iterated. Once only V_d remains, its distinct K weights and nonzero adjacent F arrows force the cut operator to be scalar.

In particular the enhanced partial traces of the crossing are

$$\mathrm{Tr}_2(B^{\pm1}(I\otimes K)) = \alpha^{\pm1} I, \qquad \alpha = q^{(N^2-1)/2}, \qquad \gamma\alpha = -q = \alpha_K. \tag{10.9}$$

For the positive trace, evaluate on v_0: only the input pair $(0,0)$ contributes. For the inverse, evaluate on v_d: only (d,d) contributes. Their values are $q^{d^2/2+d}$ and its inverse, which proves the formula. These inverse calculations can also be read from

$$R^{-1} = \left(\sum_{k=0}^{d} \frac{(-1)^k q^{-k(k-1)/2}(q-q^{-1})^k}{[k]_q!} E^k\otimes F^k\right) q^{-H\otimes H/2}.$$

The finite binomial identity makes its two polynomial factors multiply to one below degree N, and $(E\otimes F)^N = 0$.

Now take the K-weighted traces of every strand of a braid closure except the chosen first one. Its cut operator C_B is scalar by the preceding argument, irrespective of the number of link components. In (10.8), all conjugations on traced factors cancel by cyclicity. The remaining conjugation cancels because C_B is scalar. Equation (10.9) cancels every writhe factor, so

$$\alpha_K^{-w} C_K = \alpha^{-w} C_B. \tag{10.10}$$

Thus the comparison is exact even when a factorial gauge is large, and it holds at even and composite levels.

Finally change from the K-pivot to the pivot of the marked source. At completeness

$$g = (-1)^d K, \qquad \theta = (-1)^d \alpha = q^{-d^2/2}, \qquad |\theta| = 1.$$

For an s-strand closure the $s-1$ traced factors contribute $(-1)^{d(s-1)}$, and the curl normalization contributes $(-1)^{-dw}$. If the closure has c components, its braid permutation gives $w \equiv s - c \pmod 2$. Combining these signs with (10.10) proves (10.2). The turns and marked scalar are exactly those already computed in (9.5)–(9.7). For the same framed tangle, its planar ribbon

reading and its braid reading evaluate the same specified ribbon invariant. The actual reduced cut is independent of its exposed edge by the preceding source proof.

For a knot, correcting an integer framing multiplies the raw scalar by the corresponding power of the unit θ; the component sign is one. This gives (10.3). For links with separately prescribed component framings, retain those integer twist corrections rather than replacing them by the total writhe, which also counts crossings between different components. □

The full one-cut modified convention has a further, separate normalization. If its modified dimension is defined by

$$d_N(\alpha) = -q^{N(1-N)/2}\frac{q^{\alpha+1} - q^{-\alpha-1}}{q^{N\alpha} - q^{-N\alpha}},$$

then $M = -qq^{\alpha} = e^{u}$, with the lift $\alpha = d + u/\log q$, gives the whole germ

$$\begin{aligned} d_N\left(d + \frac{u}{\log q}\right) &= i^{N+3}\frac{\sinh u}{\sinh Nu}, \\ \lim_{u\to 0} d_N\left(d + \frac{u}{\log q}\right) &= \frac{i^{N+3}}{N}. \end{aligned} \tag{10.11}$$

The value at $u = 0$ is removable. It is attached once, to the selected cut component, and not to every auxiliary circle. This alternative convention is not used to change the normalization in (10.3).

11 Application after a shared coefficient extraction

Let $L_1, \dots, L_s$ be auxiliary variables and let $\mathcal{C}$ be a fixed coefficient extraction in those variables, possibly after some have been set to zero. Suppose the local spectral factors $P_e(L; \sigma_e)$ have degree at most d in σ_e. Then

$$S_N(\boldsymbol{\sigma}) = \mathcal{C}\left(\prod_e P_e(L; \sigma_e)\right)$$

has the same separate degree bounds. Theorem 1.1 applies to this shared polynomial itself. It does not replace the extraction by independent maxima of its local factors.

Corollary 11.1 (Exact-source implication)**.** *Assume a scalar contraction $Z_N(\boldsymbol{n})$ has the exact identity*

$$Z_N(\boldsymbol{n}) = \left(\bigotimes_e \mathcal{L}_N^{n_e}\right) S_N. \tag{11.1}$$

Let C_N be any other shared polynomial with the same separate degree bounds, and let $Z_{N,C} = (\bigotimes_e \mathcal{L}_N^{n_e})C_N$. Then

$$|Z_N - Z_{N,C}| \le \prod_e (1 + |n_e|/2)\, \|S_N - C_N\|_{[-2,2]^r}.$$

If $S_N - C_N$ vanishes on the product node grid, the smaller constant in (1.4) applies.

Proof. The finite functional is linear. Subtract (11.1) and the definition of $Z_{N,C}$ before applying (1.3) with $J = \varnothing$, or (1.4). This preserves the full difference, including every source sector and all shared coefficient demands. □

For the intended Morse-diagram application, the local factor is the homogeneous representation of

$$V_N(\sigma;\Omega) = \frac{\chi_N(\sigma) - \chi_N(\Omega)}{\sigma - \Omega},$$

where the fraction is polynomial division. The Fourier calculation in Section 4 proves the local adapter and the annihilator needed by resolvent reproduction. Proposition 5.1 also identifies the defined finite crossing, including its repeated-root derivatives. For the matched spherical diagrams of Section 9, Corollary 9.2 proves (11.1) for the original marked contraction. Theorem 10.1 then gives its reduced Kashaev normalization when the main link is a knot. The conditional formulation of Corollary 11.1 also permits its use with other sources whose exact evaluation has been established separately.

Here is the resulting asymptotic implication, recorded to make clear which estimate an application must supply. It is an elementary consequence of a strict remainder gap, not an additional source theorem.

Proposition 11.2 (Transfer of a strict remainder gap)**.** *Fix the number of variables and the integer tuple* $\boldsymbol{n}$*. Suppose* (11.1) *holds and, for some real* Φ *and* $\eta > 0$*,*

$$\|S_N - C_N\|_{[-2,2]^r} \le \exp\big(N(\Phi - \eta) + o(N)\big).$$

Suppose separately that

$$Z_{N,C} = a_N e^{NA}, \qquad a_N \neq 0 \text{ eventually}, \qquad \frac{1}{N}\log|a_N| \longrightarrow 0, \qquad \operatorname{Re} A > \Phi - \eta.$$

Then Z_N *is eventually nonzero and*

$$\frac{1}{N}\log|Z_N| \longrightarrow \operatorname{Re} A.$$

Multiplication by a specified scalar sequence u_N *with* $u_N \neq 0$ *eventually and* $N^{-1}\log|u_N| \to 0$ *preserves the conclusion.*

Proof. The filling constant in Corollary 11.1 is independent of N. Let $b = \Phi - \eta$ and choose $\epsilon > 0$ with $b + 2\epsilon < \operatorname{Re} A$. For all sufficiently large N,

$$|Z_N - Z_{N,C}| \le e^{N(b+\epsilon)}, \qquad |Z_{N,C}| \ge e^{N(\operatorname{Re} A - \epsilon)}.$$

Their ratio is at most $e^{-N(\operatorname{Re} A - b - 2\epsilon)}$. In particular it is eventually below $1/2$. The triangle and reverse triangle inequalities give

$$\tfrac{1}{2}|Z_{N,C}| \le |Z_N| \le \tfrac{3}{2}|Z_{N,C}|.$$

Taking logarithms and dividing by N proves the limit and eventual nonvanishing. The last assertion follows by adding $N^{-1}\log|u_N|$. □

A bound for the full source at the larger rate Φ would not satisfy this proposition when $\operatorname{Re} A < \Phi$. Nor does a leading term for C_N estimate the omitted source. These two inputs have to be proved separately for the same fixed diagram and normalization. Identifying $2\pi \operatorname{Re} A$ with a particular hyperbolic volume requires a further geometric argument; it is not part of the finite functional theorem.

References

[1] C. Hermite, *Sur la formule d'interpolation de Lagrange (extrait d'une lettre à M. Borchardt)*, Journal für die reine und angewandte Mathematik **84** (1878), 70–79. doi:10.1515/crelle-1878-18788405.

[2] M. Riesz, *Eine trigonometrische Interpolationsformel und einige Ungleichungen für Polynome*, Jahresbericht der Deutschen Mathematiker-Vereinigung **23** (1914), 354–368. Digitized original volume.

[3] R. M. Kashaev, *The hyperbolic volume of knots from the quantum dilogarithm*, Letters in Mathematical Physics **39** (1997), 269–275. doi:10.1023/A:1007364912784.

[4] H. Murakami and J. Murakami, *The colored Jones polynomials and the simplicial volume of a knot*, Acta Mathematica **186** (2001), 85–104. doi:10.1007/BF02392716.